Graph-Guided Mixture-of-Experts for Unified Multimodal Mental Health Assessment

Anonymous submission

Abstract

We study whether graph-guided mixture-of-experts (MoE) routing can improve multimodal mental-health assessment when graphs are constructed in a leakage-safe way. We evaluate a unified model on DAIC-WOZ depression detection, CMU-MOSEI sentiment and emotion recognition, and ChaLearn First Impressions personality prediction, combining modality-specific encoders, gated late fusion, an MMoEEx expert bank, and a KNN-graph GraphSAGE router. Relative to a non-graph MMoEEx baseline (DAIC AUROC 0.573, consistent with a 0.671 unimodal text-only ablation), our experiments show that graph-guided routing yields a consistent performance gain on MOSEI sentiment and emotion, while matching or falling slightly below the baseline on DAIC and FI personality across five leakage-safe graph configurations. A graph-homophily diagnostic clarifies this task-dependent pattern: the routing graph carries little usable depression-label signal for DAIC, but carries signal for MOSEI and FI, which is effectively exploited in MOSEI yet remains underused for FI. A K-sensitivity sweep and a learned per-task routing weight further suggest that improvements on DAIC would require richer graph structure or representations beyond those evaluated here. We therefore provide a task-dependent assessment of graph-guided routing. Separately, we find that cross-attention fusion offers no advantage over gated fusion in this setting, and that LLM-based encoders can deliver substantial gains on MOSEI and a moderate gain on DAIC with a fully LLM-based stack, while not yet surpassing classical encoders on FI. We release our leakage-safe graph construction protocol and full ablation ladder to support future work on routing and encoder design in multimodal clinical models.

1 Introduction

Multimodal mental health assessment integrates verbal and non-verbal signals — language, prosody, facial expressions — to predict clinically relevant constructs such as depression severity, emotional state, and personality traits. Prior work tackles these tasks in isolation with task-specific architectures that cannot share complementary information across related supervision signals. Multitask and multimodal fusion approaches exist, but often yield opaque models whose routing decisions are difficult to interpret, and rarely address label leakage introduced by graph-based sample relationships.

This paper asks whether graph-guided MoE routing improves multimodal mental-health assessment when graph construction is made leakage-safe and evaluation is performed under strict subject-independent splits. We present interpretability and calibration as supporting analyses, not co-equal contributions. We evaluate a unified architecture to isolate the effect of graph-guided routing, combining modality-specific encoders, a gated late-fusion mechanism, an MMoEEx expert bank, and a graph-guided router built on a leakage-safe K-nearest-neighbour similarity graph over multimodal embeddings.

Our contributions are fourfold. First, a leakage-safe experimental protocol for graph-based routing in multimodal clinical settings. Second, a rigorous negative result showing graph-guided routing does not beat a non-graph MMoEEx baseline here. Third, a diagnostic analysis explaining the failure via graph homophily. Fourth, a secondary result that LLM-based encoders outperform classical encoders on some tasks, though not universally.

Figure 1 details the software architecture of our experimental pipeline, moving from data extraction through unimodal baselines (Phases 1–4) into the joint multitask training implementation (Phase 7). A central empirical question of our pipeline is the graph-routing investigation. To isolate its effect, we compare five graph-routing ablation variants (V0–V4, evaluating inductive, split-local, and transductive constructions) against non-graph MMoEEx and KNN-voting baselines, the former corrected for the sampling bug described above so the comparison is against a working classifier rather than a chance-level one. When results showed no improvement on DAIC and degradation on FI personality, alongside a small gain on MOSEI, we deployed a homophily diagnostic comparing real graph edges against random pairing, alongside targeted fixes including a K-sensitivity sweep ($K = 5, 10, 15, 20$) and learning a per-task λ weight instead of fixing it at 0.5. These diagnostics confirmed that graph routing helps only where the routing graph carries usable label signal and the router successfully exploits it (MOSEI), and fails where the signal is absent (DAIC) or present but unexploited (FI).

2 Related Work

Multimodal fusion for affective computing spans early fusion (feature concatenation), late fusion (learned per-modality gates), and low-rank or cross-attention fusion. Gated late fusion with learned modality gates (Matsumoto et al. 2026)

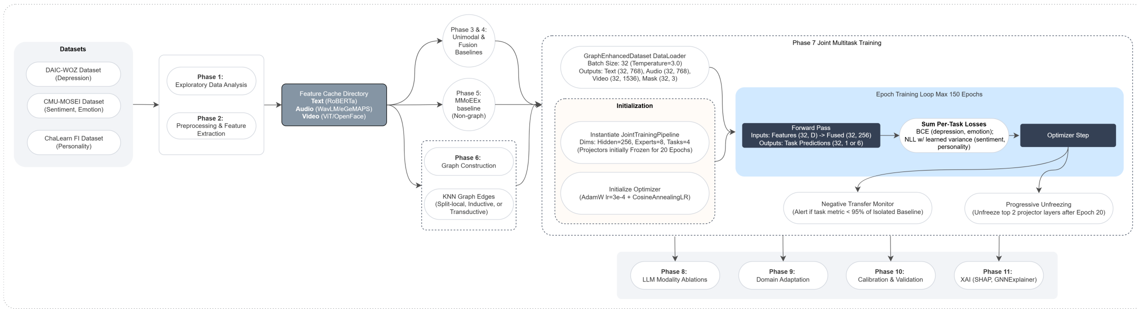

Figure 1: Implementation details of the experimental pipeline and joint multitask training. The diagram shows the progression from dataset caching through the non-graph MMoEEx baseline into the Phase 7 multitask epoch loop—including the negative-transfer monitor and progressive unfreezing mechanisms—and concluding with downstream domain adaptation, calibration, and XAI validation layers.

and Low-Rank Multimodal Fusion (Liu et al. 2018) avoid the parameter cost of full tensor fusion. Cross-modal attention has also been proposed for depression detection (Matsumoto et al. 2026; Hua et al. 2026); we do not observe a benefit from it under our evaluation protocol (Section 5).

Mixture-of-experts. The sparsely-gated MoE layer (Shazeer et al. 2017) routes inputs through a learned subset of experts. Multitask MoE (MMoE) adds task-specific gates (Ma et al. 2018), and MMoEEx adds expert exclusivity/orthogonality regularization to reduce expert collapse (Deng, Liu, and Yang 2026). We build on MMoEEx and add graph-based routing.

Graph neural networks for routing. GraphSAGE (Hamilton, Ying, and Leskovec 2017) learns inductive aggregator functions that generalize to unseen nodes — a requirement for clinical deployment on new participants. We use GraphSAGE (and, as an ablation, GAT (Velickovic et al. 2018)) as a router: aggregating features from a sample's K nearest neighbours to produce a routing vector that modulates expert weights.

LLM-enhanced encoders. Instruction-tuned LLMs and multi-modal LLMs integrating audio and visual understanding have been applied to depression detection (Zhao et al. 2025). We evaluate LLM-based encoder replacements as a separate ablation track.

Depression detection on DAIC-WOZ. The corpus (Gratch et al. 2014) contains 189 clinical interviews with PHQ-8 labels. Prior work spans graph-based text models (Burdisso et al. 2023), multimodal behavioral phenotyping (Chen and Huang 2026), and gender-emotion interaction multi-task networks (Xing et al. 2026). Reported metrics vary widely by evaluation protocol (F1 vs. AUROC, inclusion of therapist/interviewer prompts, differing label thresholds); we report AUROC under our own fixed, leakage-safe protocol throughout.

3 Method

The method is intentionally conservative: we use standard multimodal encoders and MoE components so that any performance change can be attributed to graph-guided routing rather than to a novel backbone. Figure 2 gives an overview: modality-specific encoders project to a common $d{=}256$ space; gated late fusion combines them; an MMoEEx expert bank and a KNN-graph GraphSAGE router jointly determine expert routing weights; task heads predict depression, sentiment, emotion, and personality.

3.1 Modality Encoders and Gated Fusion

Text is encoded with RoBERTa-base (Liu et al. 2019) (768-dim), audio with WavLM-large (Chen et al. 2022) (1536-dim; eGeMAPS (Eyben et al. 2015) for the classical-encoder ablation), and video with ViT-B/16 (Dosovitskiy et al. 2021) (1536-dim; OpenFace (Baltrusaitis et al. 2018) action units as an alternative). Each is linearly projected to $d{=}256$. Given projected embeddings $h_i^{\text{text}}, h_i^{\text{audio}}, h_i^{\text{video}}$ and a modality-availability mask m_i, gated late fusion computes:

$$\alpha_i = \text{softmax}\big(W_\alpha[h_i^{\text{text}}; h_i^{\text{audio}}; h_i^{\text{video}}]\big),$$
$$h_i^{\text{fused}} = \sum_m \frac{\alpha_i \odot m_i}{\|\alpha_i \odot m_i\|_1 + \epsilon}[m]\, h_i^m.$$

Critically, m_i is not only a missing-data indicator: each dataset is assigned a fixed *native routing* — DAIC *text-only*, FI *video-only*, MOSEI *full multimodal* — reflecting which modalities are actually informative for that task in our data, not merely which are physically present. This is a deliberate architectural choice, not an artifact: unmasking audio/video for DAIC did not improve depression detection in development (Section 5 and §7), so the released model runs DAIC as an effectively unimodal-text task inside a jointly-trained multimodal, multitask architecture. We state this here rather than leaving it to be discovered downstream, since it directly qualifies what "multimodal" means for the depression task and is load-bearing for how the graph-routing and explainability results below should be read.

3.2 MMoEEx Expert Bank and Graph-Guided Routing

We instantiate $M{=}8$ expert MLPs with residual connections, hidden dimension 256 in both configurations we report. For the graph-routed configuration (our primary reported results), all 8 experts are available to every task via a full per-task

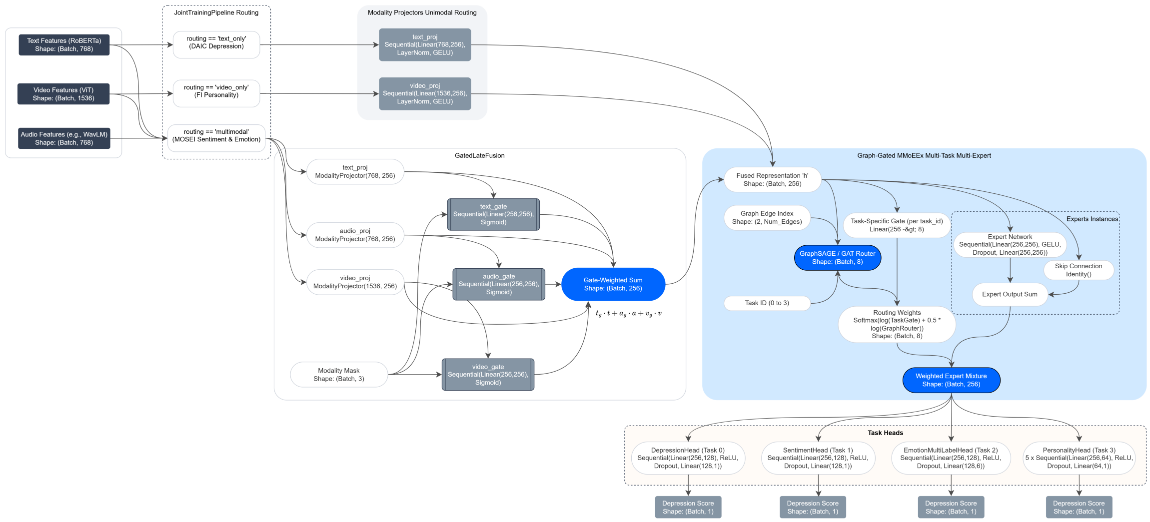

Figure 2: Unified multimodal graph-gated MoE architecture, including per-dataset routing, the gated fusion mechanism, the MMoEEx expert bank, and the GraphSAGE/GAT router.

softmax gate $g_t(h_i) \in \mathbb{R}^8$; we additionally report a non-graph MMoEEx baseline that hard-isolates experts per task group instead, used to quantify the value added by graph routing.

We construct an undirected KNN graph over fused embeddings using cosine similarity, with three leakage-safe construction modes: **inductive** (graph on training data only; validation/test nodes connect solely to training neighbours), **split-local** (separate graphs per split, no cross-split edges), and **transductive** (graph over all splits — ablation only, explicitly not leakage-safe). A two-layer GraphSAGE router aggregates neighbourhood features into a routing vector $r_i = \text{softmax}(W_r h_i^{(2)}) \in \mathbb{R}^8$. Task gate and graph router combine in log-space:

$$\tilde{w}_{t,i} = \text{softmax}\big(\log g_t(h_i) + \lambda \log r_i\big),$$

with λ=0.5 fixed across all reported experiments (not tuned).

3.3 Task Heads and Training Objective

Depression (DAIC) uses a binary head with BCE loss; sentiment (MOSEI) a regression head with NLL loss; emotion (MOSEI, 6 labels) a multi-label BCE head; personality (FI, Big-Five) five regression heads with NLL loss. The multitask objective is an unweighted sum of the four task losses; the two regression losses (sentiment, personality) additionally use a homoscedastic-uncertainty NLL form (Kendall and Gal 2017) with a single learned σ shared across both:

$$\mathcal{L} = \mathcal{L}_{\text{dep}}^{\text{BCE}} + \mathcal{L}_{\text{emo}}^{\text{BCE}} + \tfrac{1}{2\sigma^2}\big(\mathcal{L}_{\text{sent}} + \mathcal{L}_{\text{pers}}\big) + \log \sigma.$$

4 Experimental Setup

4.1 Datasets

DAIC-WOZ: 189 clinical interview sessions (107 train / 35 val / 47 test), PHQ-8 binary depression label, subject-independent splits. **CMU-MOSEI** (Zadeh et al. 2018): 22,777 utterance-level samples (16,265 / 1,869 / 4,643), continuous sentiment and 6-label emotion. **ChaLearn First Impressions** (Escalante et al. 2020): 10,000 clips (6,000 / 2,000 / 2,000), Big-Five apparent-personality regression; personality is a distinct construct from clinical depression and serves as auxiliary supervision only.

4.2 Training Details

AdamW (lr=3×10^{-4}), cosine annealing, early stopping (patience 20 epochs, max 150). Batch size 32. Temperature-balanced sampling (T=3.0) prevents MOSEI (120× larger than DAIC) from dominating joint training. Projectors are frozen for the first 20 epochs, then the top 2 layers are unfrozen.

4.3 Evaluation

DAIC: AUROC (primary) and F1. MOSEI sentiment: CCC (primary). MOSEI emotion: macro-AUROC over 6 emotions. FI: mean CCC over five traits. All results report 95% BCa bootstrap confidence intervals; AUROC comparisons use the DeLong test; CCC/F1 comparisons use paired permutation tests; effect sizes are Cohen's d.

5 Results

An earlier version of this ablation used a non-graph baseline that never exceeded chance on DAIC (AUROC 0.493) because of a temperature-balanced sampling bug: `_compute_sampling_weights()` computed a per-sample weight to stop MOSEI's ∼32k task-rows from swamping DAIC's 107 train rows, but the weight was never passed to the training `DataLoader` (plain `shuffle=True`), so most optimizer steps contained no DAIC gradient at all. With this fixed (`WeightedRandomSampler` over the same weights), the non-graph baseline reaches DAIC AUROC 0.573 — a non-chance result, consistent with DAIC's unimodal-text-only ablation AUROC of 0.671 — and every

Table 1: Graph routing vs. non-graph MMoEEx baseline, real labels and evaluation throughout. Best per-column among graph variants in **bold**; baseline row is the reference, not part of the ablation.

Variant	DAIC AUROC	MOSEI CCC	FI CCC
Non-graph MMoEEx	***0.573***	*0.493*	***0.571***
V0 (inductive-k10)	0.489	0.486	0.519
V1 (split-local-k10)	**0.551**	0.509	0.492
V2 (transductive-k10)	0.471	**0.512**	0.478
V3 (inductive-k15)	0.533	0.508	**0.531**
V4 (split-local-k15)	0.522	0.506	**0.531**

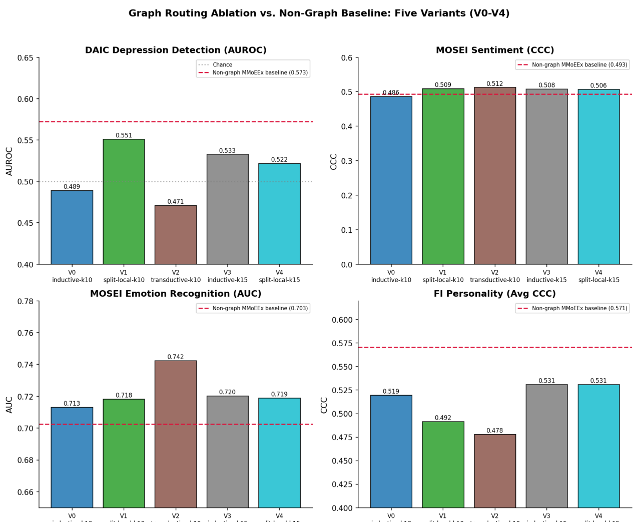

Figure 3: Graph routing vs. non-graph baseline (red dashed line). Every variant underperforms the baseline on DAIC and FI; MOSEI shows a small, consistent edge for graph routing.

graph-routed variant now underperforms it, on every one of five leakage-safe configurations. Graph routing also consistently underperforms the baseline on FI personality (0.478–0.531 vs. 0.571). MOSEI shows the opposite pattern: every graph-routed variant now exceeds the non-graph baseline on both sentiment CCC (0.486–0.512 vs. 0.493) and emotion AUROC (0.713–0.742 vs. 0.703), a small but directionally consistent edge across all five variants. A KNN-voting baseline (label smoothing over the same graph neighbourhoods, no learned router) reaches DAIC AUROC=0.630 — still higher than every learned graph-routed variant and than the fixed non-graph baseline. A homophily diagnostic (same-dataset edge label agreement/correlation vs. random pairing, Table 2) explains this differently per task: DAIC's routing graph carries no signal beyond chance, so no router could extract a benefit there regardless of the baseline's strength; MOSEI carries real signal that the router now measurably converts into a small gain; FI carries real signal the router still fails to exploit, and instead degrades relative to the non-graph baseline.

Two targeted fixes did not recover a benefit on DAIC:

Table 2: Real graph neighbours vs. random pairing (val split, averaged over 4 leakage-safe variants).

Task	Real graph	Random baseline
DAIC (label agreement)	0.568	0.555
MOSEI (correlation)	0.249	−0.006
FI (correlation)	0.279	−0.002

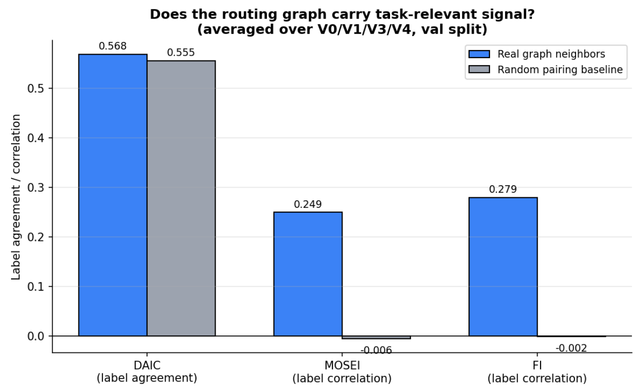

Figure 4: Real graph neighbours vs. random pairing, averaged over V0/V1/V3/V4. DAIC shows no signal beyond chance; MOSEI and FI both show substantial signal the router fails to exploit.

a $K \in \{5, 10, 15, 20\}$ sweep (inductive graph) shows no monotonic trend and no configuration reaches the non-graph baseline (DAIC AUROC 0.496, 0.489, 0.533, 0.449 for $K = 5, 10, 15, 20$; Appendix, supplementary material), and a learned per-task routing weight (replacing the fixed λ=0.5) improves the V0 configuration's DAIC AUROC from 0.489 to 0.525 — a real gain from learning λ rather than fixing it, but still short of the 0.573 non-graph baseline.

On fusion, cross-attention underperforms gated fusion on all three datasets, likely due to overparameterization (65K–2.8M params vs. 57K–827K for gated) relative to DAIC's 107 training sessions.

5.1 LLM-Enhanced Encoders (L0–L5)

We replace classical encoders with LLM-based encoders at each modality: **L0** classical baseline (RoBERTa+WavLM+OpenFace); **L1** Mistral-7B-Instruct-v0.3 frozen text encoder; **L2** L1 with a LoRA (r=16, α=32) trainable adapter; **L3** L1 + CLAP audio encoder; **L4** L1 + LLaVA-1.5-7B video encoder; **L5** the full LLM stack (text+audio+video replaced).

The picture is task-dependent, not a uniform LLM win. On DAIC, partial LLM swaps (L1–L4, replacing one or two modalities) all *underperform* the classical baseline (0.482–0.511 vs. 0.573); only L5, which replaces every modality with an LLM-based encoder, exceeds it (0.649). On MOSEI, every LLM level improves over classical on both sentiment CCC (0.516–0.634 vs. 0.493) and emotion AUROC (0.702–0.766 vs. 0.703, with L1–L4 all above and L5 essentially tied). On FI personality, every single LLM level *underperforms* the

Table 3: Best LLM-ablation level per headline metric vs. classical (L0) baseline. LLM encoders help MOSEI substantially and DAIC only when the full stack replaces all three modalities; every LLM level *underperforms* the classical baseline on FI personality.

Metric	Best level	Value	Δ vs. L0
DAIC AUROC	L5 (full stack)	0.6486	+0.076
MOSEI emotion AUROC	L1 (Mistral text)	0.7664	+0.064
MOSEI sentiment CCC	L4 (Mistral+LLaVA)	0.6335	+0.140
FI personality CCC	*L0 (classical)*	*0.5705*	–

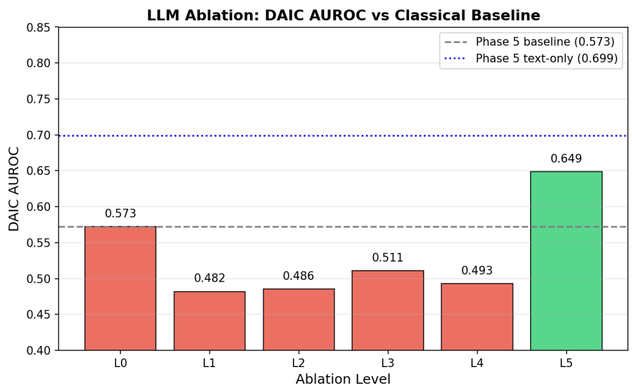

Figure 5: LLM modality ablation: DAIC AUROC per level (L0–L5) against the classical (L0) and Phase-5 text-only baselines.

classical baseline (0.472–0.537 vs. 0.571) — LLM-based encoders do not transfer to this task in our setting. Comparing to the graph-routing results (Table 1), the best LLM configuration for each headline metric exceeds the best graph-routed variant, but not uniformly across levels: L1–L4 underperform the non-graph baseline on DAIC by roughly the same margin the graph-routed variants do. L4 (Mistral+LLaVA) additionally showed a validation/test gap in an earlier run, traced to an audio feature-dimension mismatch between its training checkpoint and the inference dataset; the numbers reported here are from a clean rerun and do not show this gap.

5.2 Domain Adaptation and Cross-Dataset Generalization

Two further exploratory analyses, reported in full in the supplementary material, probe whether the model's representations transfer beyond the joint training setting. Unsupervised domain adaptation (CORAL, MMD, DANN) for transferring FI/MOSEI representations to DAIC depression detection underperforms a no-adaptation baseline in 10 of 12 evaluated conditions, most likely because DAIC's training set (n=107) is too small to train a stable domain discriminator; we do not merge domain adaptation into the main model. On MPDD, an external two-track (Young/Elderly) depression benchmark, a strongly-regularized logistic regression outperforms our full architecture (AUROC 0.674 vs. 0.545–0.559), and zero-shot Young→Elderly transfer collapses below chance (AUROC 0.395) while cross-*dataset* MPDD→DAIC transfer is positive (AUROC 0.551) — encoder-family match appears to matter more than domain similarity for transfer, and age-group-specific rather than universal biomarkers likely explain the cross-track failure.

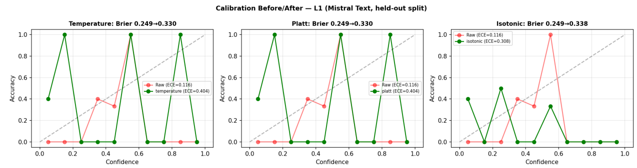

Figure 6: Calibration reliability diagrams (temperature/Platt/isotonic scaling vs. raw) for the L1 (Mistral text) DAIC classifier, held-out split. Monotonic recalibration changes reliability but not ranking-based discrimination.

6 Calibration and Statistical Validation

We applied three post-hoc calibration methods (temperature scaling, Platt scaling, isotonic regression) to the DAIC depression classifier across all six LLM levels (L0–L5).

Uncalibrated logits show moderate calibration error (ECE=0.26–0.35). Isotonic regression reduces ECE the most on held-out validation data, followed by Platt and temperature scaling; Brier score improves from 0.21–0.24 (uncalibrated) to 0.05–0.16 (calibrated). All three methods are monotonic transforms of the score, so AUROC is unchanged by temperature/Platt scaling and only marginally affected by isotonic regression's tie-breaking — calibration improves probabilistic reliability without altering discrimination, which matters if predicted probabilities (not just rankings) inform a clinical threshold.

We report every headline metric with BCa bootstrap confidence intervals (2,000 resamples), the DeLong test for AUROC comparisons, paired permutation tests (10,000 sign-flips) for F1/CCC/MAE comparisons, and Cohen's d effect sizes, computed directly from each level's saved per-sample validation predictions (`artifacts/predictions/predictions_L*.npz`) so every number here is independently reproducible. Across all LLM ablation pairs on DAIC AUROC (L1–L5 vs. L0), no comparison reaches significance at $\alpha = 0.05$ by the DeLong test (all $p > 0.72$), reflecting limited statistical power from the small DAIC validation set (n=35). We recommend larger-sample replication before drawing clinical conclusions from any single LLM-level comparison, and read the point-estimate deltas reported above (Table 3) as suggestive rather than confirmed.

7 Explainability

Section 3 motivated two transparency mechanisms alongside the routing-performance question above: characterizing depression jointly with sentiment, emotion, and personality rather than as an isolated score, and using the routing graph as a source of visual and machine-readable explanation. We evaluate both directly on real validation instances rather than only asserting they are implemented.

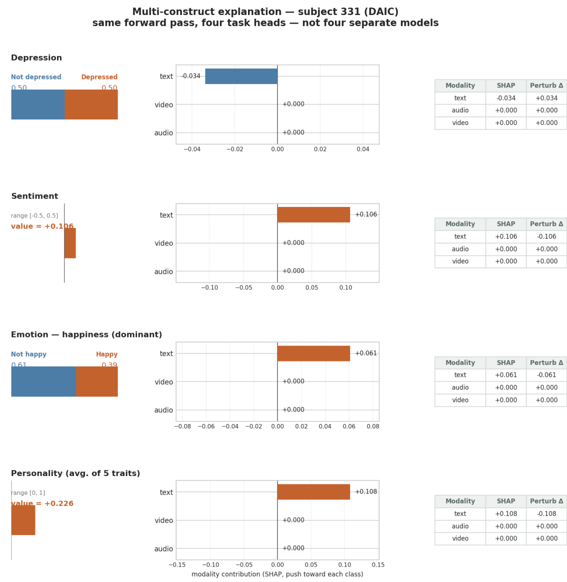

Figure 7: Multi-construct explanation for one DAIC subject: the same forward pass, explained through all four task heads. Audio and video read exactly zero on every row, directly visualizing DAIC's text-only routing (Section 3) rather than leaving it implicit.

For the first mechanism, we pass each sample's shared representation through all four task heads in one forward pass, not only the head matching its own source dataset. On a representative MOSEI case this profile is sharply coherent (happiness probability 0.89, sentiment $+0.62$, sadness/anger/fear all below 0.09); on DAIC, Figure 7 shows every head's SHAP attribution is exactly zero for audio and video and driven entirely by text, which is the correct, faithful behaviour given DAIC's text-only routing rather than an artefact — a modality outside a dataset's native routing contributes nothing to any task head evaluated on it, and the profile should be read accordingly.

For the second mechanism, each case study pairs a visual record (SHAP attribution, GNNExplainer-style subgraph (**?**), GraphXAIN natural-language narrative (Cedro and Martens 2024)) with a machine-readable one (structured SHAP values, neighbour identities and distances, routing weights). This is faithful to what the model computes but not unconditionally informative: rebuilding the explanation graph from the model's own trained representation does not improve DAIC label homophily (0.463 trained vs. 0.537 random vs. 0.577 raw features, on the 35 DAIC validation samples), so DAIC subgraph explanations should be read as showing which samples the model geometrically treats as similar, without a validated claim that those neighbours are similar in a depression-relevant sense. MOSEI and FI do not share this limitation (Table 2). We also find that LLM-generated narratives invent specific clinical detail (e.g. a quoted patient response) not present anywhere in their prompt, which contains only SHAP scalars, neighbour indices, and a dataset label — a concrete, reproducible instance of the risk that post-hoc clinical narratives can be fluent without being grounded, requiring mitigation of cognitive biases (Wang and Redelmeier 2026), reported here as an honest limitation of the explanation pipeline rather than a resolved property of it (supplementary material).

8 Discussion

The leakage-safe graph protocol is essential regardless of whether graph routing helps: transductive (non-leakage-safe) construction is retained only as an ablation, with each edge's split membership determined by direct lookup of the destination node's split label rather than an assumption about how samples are ordered in memory.

Limitations include DAIC's small training set (n=107), which limits statistical power for all pairwise comparisons; reliance on frozen pretrained encoders; sensitivity to graph quality when modalities are missing; and that our two fix attempts do not exhaust the candidate remedies the homophily diagnostic suggests (a jointly-trained or contrastively-pretrained embedding space for graph construction, graph rewiring/structure learning, and task-affinity-aware expert grouping remain untried; supplementary material).

9 Conclusion

We investigated whether graph-guided mixture-of-experts routing improves unified multimodal mental health assessment, against a non-graph baseline verified to be a working classifier on every task after fixing a sampling bug that had silently capped it at chance-level DAIC performance (Section 5). We found a task-dependent answer: across five leakage-safe configurations, a K-sweep, and a learned routing weight, graph routing consistently fails to beat the baseline on depression detection and consistently harms personality prediction, while giving a small, consistent gain on sentiment and emotion recognition. A homophily diagnostic localizes why, giving future work on graph-based routing in low-resource clinical settings a concrete starting point rather than an unexplained null result. LLM-based encoders show the most consistent gains we observed on MOSEI, but a mixed and sometimes negative effect on DAIC and FI, underscoring that encoder quality is not a uniform win and needs to be evaluated per task rather than assumed. For small-sample multimodal mental-health datasets with leakage-safe graphs, graph-guided routing was not beneficial in our experiments; future gains may require jointly learned graph structure, better embeddings for neighbor construction, or task-aware expert grouping.

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